

1 Configuration for the Expanded View of Quality Data

	• →	
--	-----	--

Purpose

Use this function, if you want to show further data in addition to the input dialog in the AIP inspection data collection. For example, you can show the list of documents or directly show a drawing.

The expanded view of quality data is based on a customizable layout for the inspection data collection.

Requirements

The functions described below are available for HYDRA 8 with AIP 8.2 service pack 10 and higher. You also need the license AIP-EQD.

For final activation, set the following parameter in the CAQ configuration file *caq_dc_t.ini* (folder: *functions*) in the *[OPTIONS]* section:

```
ACTIVATE_EQD=1
```

Basic configuration

Without the expanded view of quality data, the AIP inspection data collection always shows both: the inspection list and the dialog pertaining to the entry selected in the inspection list. The inspection list is always on the left-hand side. The relevant input dialog is always to the right of it. Use the expanded view of quality data to arrange these two objects individually. In addition to this, you can also show further quality data simultaneously. You can also arrange these additional objects according to your requirements.

Define the corresponding basic configuration in the file *caq_global_gui_settings.xml*. This file is stored in the AIP sub-folder *.\functions*.

Definition of grids

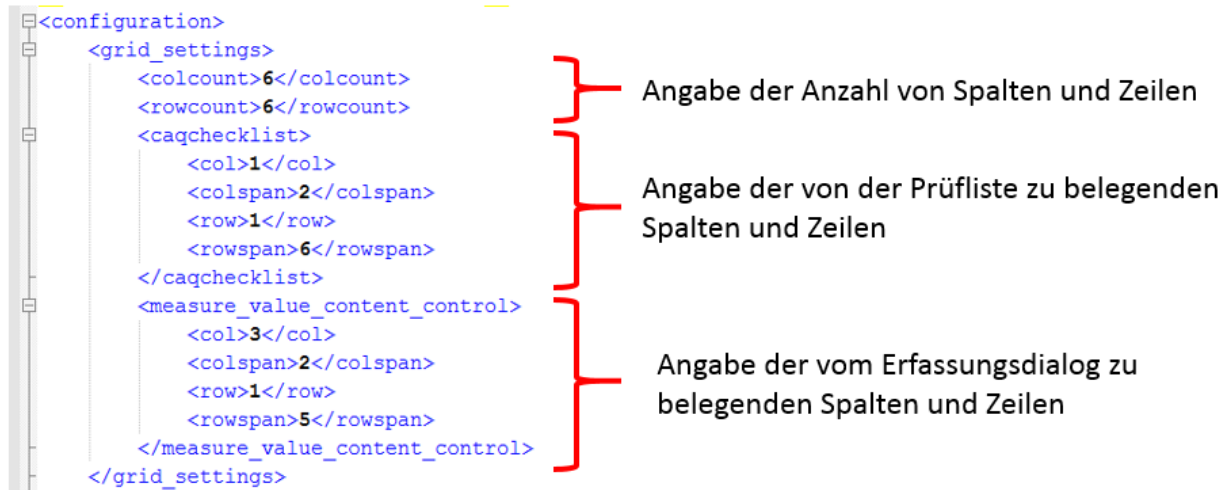
In the *<grid_settings>* section of this file, first divide the screen area into the required number of columns and rows.

Layout of inspection lists and dialogs

After the grids have been defined, specify the required columns and rows in the section *<caqchecklist>* to define which cells are populated by the inspection list. Define the display area for the input dialog in the *<measure_value_content_control>* section.

Use the parameters *col* (columns) and *row* (rows) to define where the cells that are populated begin. The *colspan* and *rowspan* parameters define the number of columns and rows that are populated. If the column and row width equals 1, you do not have to specify the *span* parameter.

The following illustration shows a configuration example.



The example configuration above would result in the following layout for inspection list and input dialogs.

	Spalte 1	Spalte 2	Spalte 3	Spalte 4	Spalte 5	Spalte 6
Zeile 1	Anzeigebereich für die Prüfliste		Anzeigebereich für die Erfassungsdialoge			
Zeile 2						
Zeile 3						
Zeile 4						
Zeile 5						
Zeile 6						

Context elements

You can use the section `<context_elements>` to define context elements with a different database. For example, you require the definition of context elements to specify which change of a node or element does not trigger an update of the contents of the objects displayed additionally in the inspection list. For example, the document list or other objects need usually not be updated when you change from measured values container 1 to the next measured values container within the same characteristic. If you suppress an update, you can enter further data within a shorter time. Therefore, we recommend this data suppression.

To define a context element, make the following entry in the `<context_elements>` section.

```
<context_element id="context name">
```

The context name can be any name. Below the context element, define the parameters that clearly describe this context. With regard to the example about updating above, this means that the system triggers an update anytime the content of these parameters changes. Define the context parameters as follows:

```
<fields>Parameter1|Parameter2|Parameter3</fields>
```

The following parameters are available, among others.

- RECTYP e.g. WEP, FEP, WAP
- BER e.g. Areas of the inspection requirement
- PANNR Inspection requirement number
- PAUNR Inspection step number
- AFO Operation sequence number
- WERTNR Measured value number
- STPRNR Sample number
- EINTNR Inspection point number

Use a "|" to separate multiple parameters.

Below is an example of a configuration for the contexts *NO_RELOAD**, *characteristic*, *measured_value*, *inspection_point* and *sample*.

```

<context_elements>
  <context_element id="static">
    <fields>NO_RELOAD</fields>
  </context_element>
  <context_element id="inspection_step">
    <fields>RECTYP|BER|PANNR|PAUNR</fields>
  </context_element>
  <context_element id="characteristic">
    <fields>RECTYP|BER|PANNR|PAUNR|AFO</fields>
  </context_element>
  <context_element id="measure_value">
    <fields>RECTYP|BER|PANNR|PAUNR|AFO|WERTNR</fields>
  </context_element>
  <context_element id="inspection_point">
    <fields>RECTYP|BER|PANNR|PAUNR|EINTNR</fields>
  </context_element>
  <context_element id="sample">
    <fields>RECTYP|BER|PANNR|PAUNR|STPRNR</fields>
  </context_element>
</context_elements>

```

The *static* = *NO_RELOAD* element is an exception. This element can be used to completely disable the update of components.

Placeholders

Using the *context_placeholder* element, you can define separate placeholders, which are then replaced with the relevant values from the CAQ node data when setting up the list parameters.

For

example:

Defining placeholders:

```

<context_placeholders>
  <placeholder>
    <name>%RECTYP%</name>
    <value>RECTYP</value>
  </placeholder>
  <placeholder>
    <name>%AFO%</name>
    <value>AFO</value>
  </placeholder>
  <placeholder>
    <name>%PANNR%</name>
    <value>PANNR</value>
  </placeholder>
  <placeholder>
    <name>%PAUNR%</name>
    <value>PAUNR</value>
  </placeholder>
  <placeholder>
    <name>%BER%</name>
    <value>BER</value>
  </placeholder>
  <placeholder>
    <name>%ATK%</name>
    <value>ATK</value>
  </placeholder>
</context_placeholders>

```

In a specified configuration, these placeholders replace the placeholder %RECTYP% with the RECTYP value from the current CAQ node data, for example.

Options

Define global settings in the <options> section. These settings serve as the default values in the following dialog configuration. If you defined settings in the "options" section, you do not have to repeat these configurations in the dialog configurations.

Currently, you can globally define the scroll mode for lists and the zoom option for documents. The available scroll modes are *scrollbars* and *buttons*.

Fehlerarten zum Merkmal	
Fehlerart	Datum
Verunreinigung	18.01.2013
Kratzer	20.01.2013
Grat	21.01.2013
Rost	26.01.2013
Haarriss	26.01.2013

Mode: *scrollbars*

Maßnahmen zum Merkmal	
Maßnahme	Datum
Teile ölen	24.05.201
Prüfung verschärfen	24.05.201

Mode: *buttons*

Use the *true/false* parameter to enable/disable the zoom function. If the zoom mode is activated, the



icon appears on the bottom right in the document/document text displayed.

To scale the measured value input dialog, use the *default_scale_factor* option to change the calculated scaling factor subsequently. 100 equals the calculated factor.



You can change the design of the inspection list to a certain extent.

If you want to change the design of the inspection list, use the parameter "[FONT_VIRTUALSTRINGTREE] → *DefaultNodeHeight*" of the configuration file *caq_dc_t.ini*.

For further information, refer to the relevant configuration description.

To request control chart and histogram data, you can use different data bases. They are configured in the following options:

- *default_controlchart_context*
- *default_histogram_context*

If you want to request a control chart for one characteristic, define the following fields:

RECTYP|BER|PANNR|PAUNR|AFO

If the control chart should be requested for an article for all orders, define the following fields:

RECTYP|BER|ATK|AFO



You can also define the data basis for each control chart or histogram in the *chart_data_source* configuration.

The following illustration shows a possible configuration:

```
<options>
  <option id="default_scale_factor">85</option>
  <option id="global_scroll_mode">buttons</option>
  <option id="can_zoom_document">true</option>
  <option id="default_controlchart_context">RECTYP|BER|PANNR|PAUNR|AFO</option>
  <option id="default_histogram_context">RECTYP|BER|PANNR|PAUNR|AFO</option>
</options>
```

Dialogs

You assign layout configurations to the relevant input dialogs in the *<dialogs>* section.

Specify the dialog as follows:

```
<dialog id="dialog specification">
```

The dialog used to collect measured values including a reference to the inspection point is, for example: *qee_mw_me_es_pp_si* and requires the following configuration. Use lower case letters for the specification.

```
<dialog id="qee_mw_me_es_pp_si">
```

The dialog specifications are included in the MOC application of the dialog configuration. The relevant input dialogs for inspection data generally start with *QEE*.



The above-mentioned configurations are always created for the first dynamic dialog of the relevant workflow configuration.

In the example above, the relevant input workflow is *MW_ME_ES_PP_SI*. Its first dynamic dialog has the name *QEE_MW_ME_ES_PP_SI*.

After the dialog has been specified, assign the layout XML file as follows:

```
<configuration_file>layout.xml</configuration_file>
```

You can define any XML file name.

```
<dialogs>
  <dialog id="qee_inspstep">
    <configuration_file>inspection_step.xml</configuration_file>
  </dialog>
  <dialog id="qee_inspoint">
    <configuration_file>insppoint.xml</configuration_file>
  </dialog>
  <dialog id="qee_sample">
    <configuration_file>insppoint.xml</configuration_file>
  </dialog>
  <dialog id="qee_mw_me_es_pp_si">
    <configuration_file>variable.xml</configuration_file>
  </dialog>
  <dialog id="qee_mm_me_es_pp_si">
    <configuration_file>characteristic.xml</configuration_file>
  </dialog>
  <dialog id="qee_mw_me_es_pp_ca">
    <configuration_file>variable.xml</configuration_file>
  </dialog>
  <dialog id="qee_mm_me_es_pp_ca">
    <configuration_file>characteristic.xml</configuration_file>
  </dialog>
  <dialog id="qee_mm_co_st_pp_si">
    <configuration_file>attributive.xml</configuration_file>
  </dialog>
  <dialog id="qee_mm_be_st_pp_fs">
    <configuration_file>inspection_chart.xml</configuration_file>
  </dialog>
</dialogs>
```

General element configurations

Configuration	Description	Valid values
context_change	Collection of acronyms causing the element to be refreshed.	The applicable values result from the definition of all context change methods of the global configuration.
workdir	Use this parameter to define a working directory in the corresponding CAQ order folder.	Use a valid folder name. If this value is not defined, the system assigns a name. The "workdir" specifications must be unambiguous within an XML file. This means that each element must have a different "workdir" specification. Static elements must have a "workdir" specification that is unambiguous with all XML files.

To display an element, you must or you can set the following attributes in the configuration for the element:

Configuration	Description
class	Defines the element class. The system currently supports the following element classes: - <i>ControlChart1</i> - <i>ControlChart2</i> - <i>Histogram</i> - <i>DocumentControl</i> - <i>Grid</i> - Custom class
col	Defines the column showing the element.
row	Defines the row showing the element.
colspan	(Optional) Defines the number of columns that should be displayed.
rowspan	(Optional) Defines the number of rows that should be displayed.

description	(Optional) Use this attribute to give the element a name. Then this name is displayed in the validation program if an error occurs.
-------------	---

Sample configuration:

```
<element class="Grid" col="5" colspan="2" row="1" rowspan="2" description="Failure types over all characteristics for the inspection step">
  <settings>
    <workdir>FailureTypes_InspectionStep</workdir>
    <context_change>inspection_step</context_change>
    <header_title></header_title>
    <header_visible>false</header_visible>
    <list_source>CPAUERR.LIST</list_source>
    <list_dynamic_filter>CPAUERR.RECTYP=%RECTYP%|CPAUERR.BER=%BER%|CPAUERR.PANNR=%PANNR%|CPAUERR.PAUNR=%PAUNR%</list_dynamic_filter>
    <list_static_parameter>CPAUERR.ERRTYP=FA</list_static_parameter>
    <columndef_section_name>QM-failure_type_EQD</columndef_section_name>
  </settings>
</element>
```

Element configurations to request data from the HYDRA server

Elements, which request data from the HYDRA server, require additional configurations. Include the following parameters for this purpose:

Configuration	Description	Valid values
list_source	Configuration of the list requested	See HYDRA lists
list_dynamic_filter	Dynamic filters to request data from the HYDRA server. The system then fills the specified acronyms with the values of the current node. e.g.: <i>RECTYP=%RECTYP%</i> <i>becomes: RECTYP=FEP</i>	All elements of the global configuration for the element <i><context_placeholder></i> .
list_static_parameter	Static parameters that are directly attached to the command for the HYDRA server. If you want to enable a combined filtering of a field content, e.g. showing failure types and failure causes in one grid, then separate them using a pipe slash (" ").	

Configuration example:

```

<element class="Grid" col="5" colspan="2" row="1" rowspan="2" description="Failure types and causes over all characteristics for the inspection point">
  <settings>
    <workdir>FailureTypes_InspectionPoint</workdir>
    <context_change>sample</context_change>
    <header_title></header_title>
    <header_visible>false</header_visible>
    <list_source>CPAUERR.LIST</list_source>
    <list_dynamic_filter>CPAUERR.RECTYP=%RECTYP%|CPAUERR.BER=%BER%|CPAUERR.PANNR=%PANNR%|CPAUERR.PAUNR=%PAUNR%|CPAUERR.STPRNR=%STPRNR%</list_dynamic_filter>
    <list_static_parameter>CPAUERR.ERRTYP=FA|FU</list_static_parameter>
    <columnider_section_name>CP-Failure-EQD</columnider_section_name>
  </settings>
</element>

```

Control chart component

The class names are *ControlChart1* and *ControlChart2*. The numbers 1 and 2 refer to the control charts 1 and 2 you can define in the MOC.

Configuration	Description	Default value*	Valid values
header_font_style	Header is shown in bold and/or italic		italic bold italic, bold
show_y_axis_desc	Shows the Y-axis	false	true false
show_x_axis_desc	Shows the X-axis	false	true false
show_specified_value	Shows the target value	false	true false
show_tolerance_limits	Shows tolerance limits	true	true false
header_visible	Shows header	false	true false
show_chart_caption	Shows the control chart description	true	true false
scale_by_y_axis	Scaling based on the Y-axis	false	true false
show_single_values	Shows each of the measured values	false	true false
show_grid_lines	Shows the grid Restriction: The grid is only shown if you enabled the display of both axis labels.	false	true false
workdir	Temporary working directory		
show_action_limits	Shows action limits	true	true false

header_title	Caption		
show_warning_limits	Shows warning limits	false	true false
measure_value_count	Number of measured values to be displayed	10	
background_color	Background color	R=255,G=255,B=255	
x_axis_field	X-axis label	Sample	All acronyms of file "RGK.lst" Special acronyms see (1) below the table
chart_data_source	Use these fields to request the relevant control chart data from the HYDRA server. If no definition is specified here, the system uses the values of the global option <default_controlchart_context>.	RECTYP BER PANNR PAUNR AFO	

type	Use this parameter if, as opposed to the control chart definition, you want to show another control chart in inspection planning. The feasible values are the active status IDs of the status type <i>REGELKART</i>. Refer to the status application to view these status IDs.		
------	--	--	--

*The default value will be used if you did not explicitly define a configuration. Therefore, you do not have to configure these properties with the default value.

(1) For the following acronyms, specially formatted outputs are available.

Acronym from RGK.Istt	Formatted acronym	Output format
WERT:DAT	DatumE	dd.mm.yyyy
WERT:ZEI	ZeitE	hh:mm
LWERT:DAT	DatumL	dd.mm.yyyy
LWERT:ZE	ZeitL	hh:mm
	Datum Zeit E Combined using DatumE + ZeitE	
WERT:VONK	Datum Zeit L Combined using DatumL + ZeitL	
SPABS:ZEI	ZeitA	hh:mm
SPABS:DAT	DatumA	dd.mm.yyyy
	Datum Zeit A Combined value from DatumL + ZeitL	

Sample configuration:

```

<element class="ControlChart1" col="3" colspan="1" row="6" description="Control chart 1">
  <settings>
    <workdir>ControlChart_1</workdir>
    <context_change>characteristic</context_change>
    <show_single_values>true</show_single_values>
    <show_specified_value>true</show_specified_value>
    <show_x_axis_desc>false</show_x_axis_desc>
    <show_y_axis_desc>true</show_y_axis_desc>
    <show_tolerance_limits>true</show_tolerance_limits>
    <show_action_limits>true</show_action_limits>
    <show_warning_limits>true</show_warning_limits>
    <show_chart_caption>true</show_chart_caption>
    <show_grid_lines>false</show_grid_lines>
    <background_color>R=213,G=224,B=245</background_color> <!--for example light blue: R=213,G=224,B=245-->
    <header_title>R</header_title>
    <header_visible>false</header_visible>
    <header_font_style>bold,italic</header_font_style>
    <measure_value_count>20</measure_value_count>
    <chart_data_source></chart_data_source>
    <type>XMED</type>
    <x_axis_field></x_axis_field>
  </settings>
</element>

```

Histogram component

Class name: Histogram

Configuration	Description	Default value*	Valid values
show_grid_lines	Shows the grid Restriction: The grid is only shown if you enabled the display of both axis labels.	false	true false
class_count	Number of displayed classes. If nothing is specified, the HYDRA server calculates the number.		
header_font_style	Header is in bold and/or italic		italic bold italic, bold
header_title	Caption		
show_y_axis_desc	Shows the Y-axis	false	true false
show_x_axis_desc	Shows the X-axis	false	true false
background_color	Background color	R=255,G=255,B=255	

show_chart_caption	Shows the control chart description	true	true false
header_visible	Shows header	false	true false
chart_data_source	Use these fields to request the relevant control chart data from the HYDRA server. If nothing is specified here, the system uses the values from the global option <default_histogram_context>.	RECTYP PANNR PAUNR AFO	

Sample configuration:

```
<element class="Histogramm" col="4" colspan="1" row="6" description="Histogram">
  <settings>
    <workdir>Histogramm</workdir>
    <context_change>characteristic</context_change>
    <show_intervention_limit>>false</show_intervention_limit>
    <show_x_axis_desc>>false</show_x_axis_desc>
    <show_y_axis_desc>>true</show_y_axis_desc>
    <show_chart_caption>>false</show_chart_caption>
    <background_color>R=213,G=224,B=245</background_color> <!--for example light blue: R=213,G=224,B=245-->
    <class_count>7</class_count>
    <chart_data_source></chart_data_source>
    <header_title></header_title>
    <header_visible>>false</header_visible>
    <show_grid_lines>>false</show_grid_lines>
    <header_font_style>bold,italic</header_font_style>
  </settings>
</element>
```

Document view showing documents directly

Class name: DocumentControl

Configuration	Description	Default value*	Valid values
document_source	Type of document view Set this value always to <i>grid</i> for this presentation component. Generally possible parameters: grid = List of documents immediately showing the selected document		grid

	list_only = List of documents without showing the selected document immediately file = Directly showing the document with a fixed position number		
header_title	Caption		
can_zoom_document	Specifies if the zoom button is shown. If this parameter is not specified, the system checks the global option <can_zoom_document>.	True	true false
header_visible	Shows header	false	true false
header_font_style	Header is in bold and/or italic		Italic bold italic, bold
entry_id	ID of the document to be shown. Refers to the <i>DOKNR</i> field in the list of documents.		
list_static_parameter	The static parameters mentioned in "List,56" in the section "list components" are available.		

With this element, you must also specify the configuration parameters to request lists from the HYDRA server.

Sample configuration:

```
<element class="DocumentControl" col="5" colspan="2" row="1" rowspan="4" description="document list with displaying a document">
  <settings>
    <workdir>documents</workdir>
    <context_change>characteristic</context_change>
    <document_source>grid</document_source>
    <header_visible>>false</header_visible>
    <view_align>vertical</view_align>
    <list_source>LIST;56</list_source>
    <list_dynamic_filter>RECTYP=%RECTYP% | BER=%BER% | PANNR=%PANNR% | PAUNR=%PAUNR% | AFO=%AFO%</list_dynamic_filter>
    <list_static_parameter></list_static_parameter>
    <columndef_section_name>QM-documents_EQD</columndef_section_name>
    <percentage_grid_usage>60</percentage_grid_usage>
    <can_zoom_document></can_zoom_document>
  </settings>
</element>
```

List of documents without showing the document

Class name: DocumentControl

Configuration	Description	Default value*	Valid values
document_source	Type of document view Set this value always to <i>list_only</i> for this presentation component. Generally possible parameters: grid = List of documents immediately showing the selected document list_only = List of documents without showing the selected document immediately file = Directly showing the document with a fixed position number		list_only
columndef_section_name	Refers to the column definition from ctaipplay.ini		
header_font_style	Header is in bold and/or italic		italic bold italic, bold
header_title	Caption		
scroll_mode	Scroll mode of the grid element	buttons	buttons scrollbars
list_grid_local_filter	Filter applied to the grid after requesting and preparing data.		
header_visible	Shows header	false	true false
list_static_parameter	The static parameters mentioned in "List,56" in the section "list components" are available.		

With this element, you must also specify the configuration parameters to request lists from the HYDRA server.

Lists with document view

Class name: DocumentControl

Configuration	Description	Default value*	Valid values
document_source	Type of document view Set this value always to <i>file</i> for this presentation component. Generally possible parameters: grid = List of documents immediately showing the selected document list_only = List of documents without showing the selected document immediately file = Directly showing the document with a fixed position number		file
columndef_section_name	Refers to the column definition from ctaipplay.ini		
percentage_grid_usage	States the percentage for zooming in/out the document grid	50	
view_align	Aligns the elements (above or below each other)	Vertical	vertical horizontal
header_font_style	Header is in bold and/or italic		italic bold italic, bold
header_title	Caption		
scroll_mode	Scroll mode of the grid element	buttons	buttons scrollbars
can_zoom_document	Specifies if the zoom button is shown.		

header_visible	Shows header	false	true false
----------------	--------------	-------	---------------

With this element, you must also specify the configuration parameters to request lists from the HYDRA server.

List components

Class name: Grid

Configuration	Description	Default value*	Valid values
columndef_section_name	Refers to the column definition from the file "ctaisplay.ini"		
list_static_parameter	Static parameters for the HYDRA server command to be executed		
header_font_style	Header is in bold and/or italic		italic bold italic, bold
header_title	Caption		
scroll_mode	Scroll mode of the grid element	buttons	buttons scrollbars
header_visible	Shows header	false	true false

With this element, you must also specify the configuration parameters to request lists from the HYDRA server.

Static elements

Static elements are configured exactly as all other available elements. The only difference is that they are configured in the global configuration file in the element *static_elements*.

In the `<static_elements>` section configure elements that should never be shown with changed content irrespective of the node. One example is the continuous display of a drawing. A drawing can be displayed unchanged irrespective of the current node. The advantage of static elements is that you only have to request these elements once. If you change nodes in the inspection list, the system does not have to request and refresh this element data once more. Below is an example for the definition of the static display of a document with item number 99:

```
<static_elements>
  <element class="DocumentControl" col="5" colspan="2" row="5" rowspan="2">
    <settings>
      <workdir>static_document</workdir>
      <document_source>file</document_source>
      <context_change>inspection_requirement</context_change>
      <header_title>Dokument feste Position</header_title>
      <header_visible>false</header_visible>
      <entry_id>99</entry_id>
      <list_source>LIST;56</list_source>
      <list_dynamic_filter>RECTYP=%RECTYP%|BER=%BER%|PANNR=%PANNR%</list_dynamic_filter>
      <columndef_section_name>QM-Merkmalsdokumente_Zusatz</columndef_section_name>
    </settings>
  </element>
</static_elements>
```

OCX extension component

An extension component can include all configurations from the *General Element Configurations*. Further options are managed within the component.

Technical implementation for customer extensions:

The component must support the following functions:

- AIPDataChanged([in] BSTR value);
 - No return value
 - Data are transmitted as a string.
 - Object Pascal (Delphi) procedure AIPDataChanged(const Value: WideString);

The data string includes the individual fields (values) separated by a pipe. The function is called up when the context changes.

Configure the context change in the corresponding element using the following XML configuration:

```
<context_change>characteristics_value</context_change>
```

- BeforeRemoveElement
 - No return value
 - No data transmission

The function is called before the element is destroyed.

Creating an OCX component in Delphi XE7

Start by setting up a new *ActiveX-Library* type project.

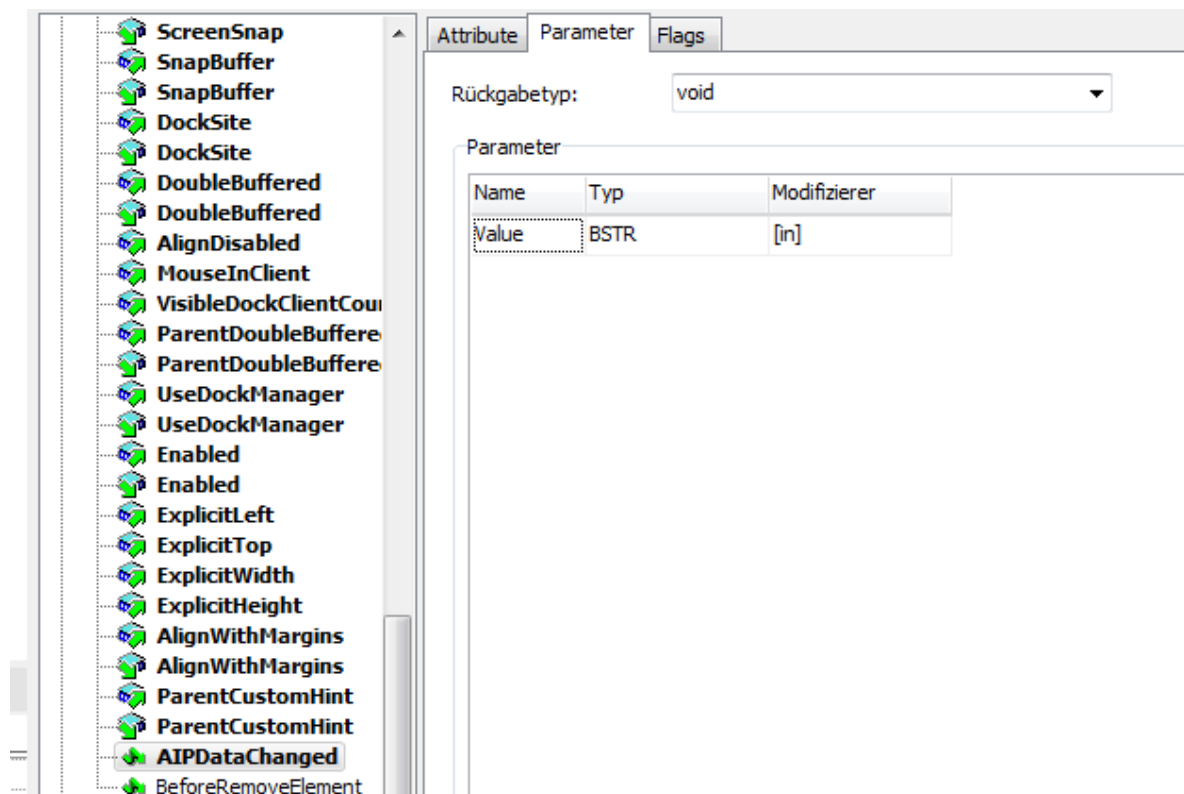
Then, store the project with a relevant name. Do this in the menu *File --> Save project as*. You are prompted to store three files.

In the next step, add an ActiveX form to the project. Go to the menu item *File|New|Other*.

Define the name of the *CoClass*. This name is equivalent to the COM object name in which the interface can be implemented.

The relevant interface is created after clicking *Confirm*. The system implements the functions in the generated form class.

Add the functions listed above. In addition, use the tab button to define the corresponding function parameters (see illustration).



Finally, update the implementation. The system adds the new functions to the form class.



An ActiveX component with the name *Project_name.ocx* is created after clicking on *Save*. Register this component as administrator with *regsvr32*.

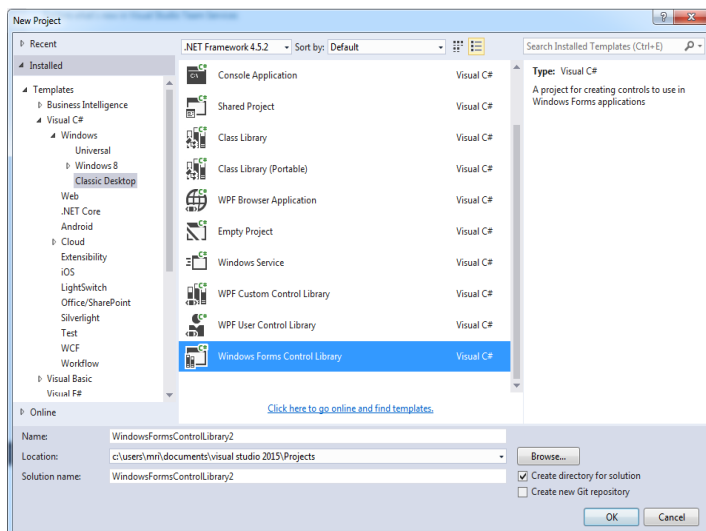
Now integrate this component as an element in the configuration for the relevant measured value dialog. The class name matches the name of the ActiveX Library and the coClass name.

In our example, it is *ExampleOCX.ExampleDisplayClass*.

```
<element class="ExampleOCX.ExampleDisplayClass" col="5" colspan="2" row="1" rowspan="2">
  <settings>
    <context_change>characteristics_value</context_change>
  </settings>
</element>
```

Creating in Visual Studio 2015

Create a project *Windows Forms Control Library*.



Add the following attributes:

- ComVisible(true)
- Guid → should be re-generated
- ProgId → name with which the COM class is accessed
- ClassInterface(ClassInterfaceType.AutoDual)

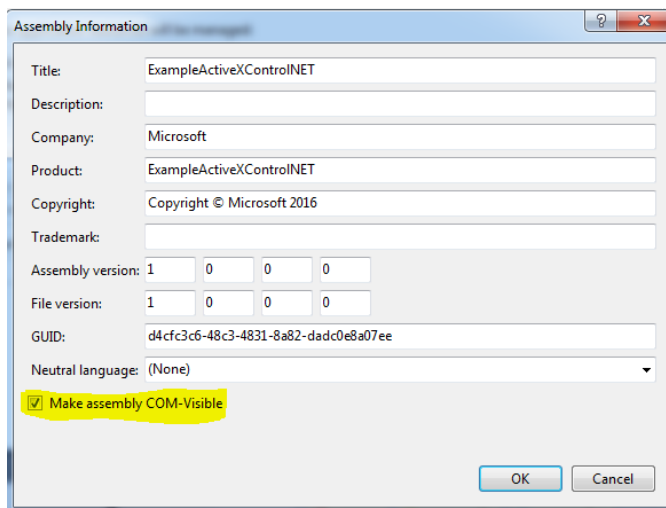
In addition, add the functions mentioned in the section "Technical implementation for customizations".

```
namespace ExampleActiveXControlNET
{
    // WICHTIG für COM:
    // ProgId ist für VC++ relevant
    // VB6 orientiert sich am Assemblynamen
    // am besten man benennt Assembly und ProgId gleich
    [ComVisible(true)]
    [Guid("1E1738B5-3C2A-4928-B029-C7F29DC8CEA0")] // GUID muss evtl. neu generieren
    [ProgId("ExampleActiveXControlNET")]
    [ClassInterface(ClassInterfaceType.AutoDual)]
    2 references
    public partial class ExampleActiveXControl: UserControl
    {
        0 references
        public ExampleActiveXControl()
        {
            InitializeComponent();
        }

        0 references
        public void AIPDataChanged(string value)
        {
            label12.Text = value;
        }

        0 references
        public void BeforeRemoveElement()
        {
        }
    }
}
```

In the next step, make the assembly COM visible in the project properties.



Finally, register the DLL with *regasm.exe*. Use the command:

```
regasm.exe /codebase mylib.dll
```



When configuring static elements, make sure that they are only used in one scope! Otherwise there may be side effects when positioning and displaying individual elements as a result of merging the global XML file.

HYDRA lists

Designation	Call parameters
Inspection order header data	List;54 Pruefschritt.lst
Characteristic data	List;55 Merkmal.lst
Further applicable documents/texts	List;56 data.lst (in the folder according to the specification in the relevant XML configuration file) The following "list_static_parameters" are available: • MOD:KONTEXT=PAN Only shows the documents of the inspection requirement. • MOD:KONTEXT=PAUMM Only shows the documents of the characteristic. Filter dynamically by the "OP sequence" and do not use the above-mentioned static parameters in order to show the documents of the characteristic and the inspection requirement.
Failure type catalog	List;57
Measures catalog	List;58
Measured values	List;60 Einzelwerte.lst
Sample status	List;65
Assessment catalogs	List;70

Control chart data	List;71 RGK.lst
List with results of the inspection order search	List;72
Histogram data	List;79 data.lst (in the folder according to the specification in the relevant XML configuration file)
INI entries from DB	List;95
List of logged on inspection orders	List;96
Statistics calculation	List;99 "AFO"_char_var_stat.lst, e.g. 100_char_var_stat.lst
Analysis selection catalogs	List;100 MerkmalAnAusKat.lst
CAQ options	List;101
Number pool	List;111
Tools	List;112
Cavities	List;122
Samples	List;123 Stichproben.lst
Inspection points	List;124 PPunkt.lst
Failures and measures relating to a machine	List;127

Validation of the configuration

You can use a program to validate the configuration changes that you have made.

The program *configurationvalidator.exe* is stored on the HYDRA server in the subfolder *.\ctnet\win\shared*.

Before using the program, make a local copy of the program on your host.

After program start, use the field *Select AIP root directory* to select the AIP directory that contains the changed configurations. Use the AIP main directory and not the subdirectory *.lfunctions*.

Select in field "Group" the entry "CAQ" and in field "Category" the entry "EQD".

Use the button *Validate configuration* to validate all included EQD configurations. The list at the bottom of the window shows the validation results.

Defective entries are identified via a red symbol. Messages have a blue symbol. Validated configuration files that do not include any messages or errors are identified via a green check.



You can ignore error messages with the additional information "-> *The defined class <???_inactive> could not be found*".

The error message refers to inactive areas that you can easily integrate into your configuration. Just remove the „_inactive“ in the class name "class=".